Quantile Regression with Censored Selection and Many Controls

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Outline

Motivation

Core Issue: Non-Random Sample Selection

- Individuals **self-select** into groups based on unobserved characteristics

Example: Wage Analysis

- Only observe wages for **employed** individuals
- Selection Bias: Unobserved factors (ability, skills) affect both employment status and wage level

Empirical Illustration

Data

American Community Survey 2001–2023

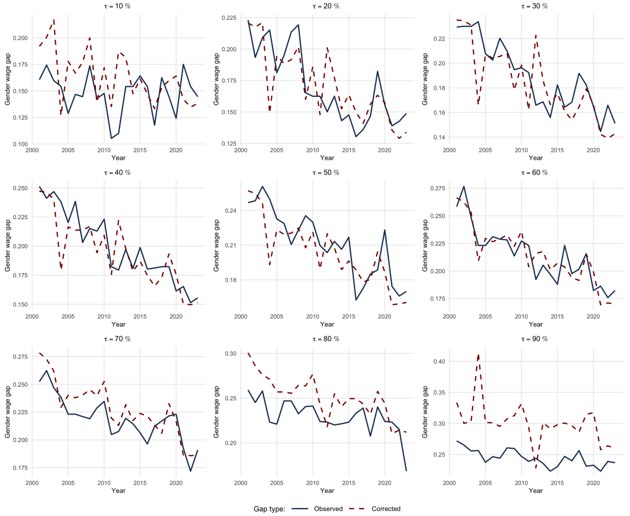
Method

Selection-corrected wage predictions

Goal: Correcting for Selection Bias

Next: results for gender wage gaps across quantiles

Key Findings



Gender wage gaps across quantiles (2001 - 2023)

Methodological Framework

Quantile Regression + LASSO
Distributional analysis + High-dimensional control

Why Quantile Regression?

- **Beyond averages**: Capture distributional effects
- **Policy relevance**: Education impacts differ across wage levels
- **Inequality focus**: Essential for wage gap analysis

Methodological Framework

Quantile Regression + LASSO
Distributional analysis + High-dimensional control

Why LASSO?

- **Necessity**: Address cases where
  number of controls > sample size
- **Flexibility**: Framework accommodates various ML methods
  Adaptive LASSO, tree-based models, etc.

Address Bias

Contribution 1: Identification

Arellano & Bonhomme (*Econometrica*, 2017)

Method of Moments (MM)

- Conditional moments

  $$E(\varepsilon|X) = 0$$

Our Approach

Nonlinear Least Square (NLS)

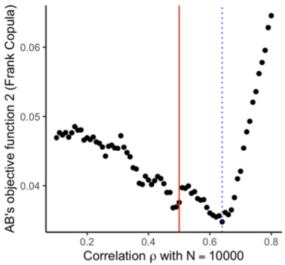
- Unconditional moments

  $$\min E(Y - m(X))^2$$

Contribution 2: Estimation

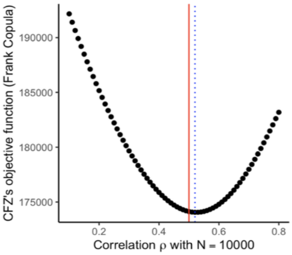
Arellano & Bonhomme (*Econometrica*, 2017)

- Instrument selection required
- Computationally unstable



Our Approach

- No instrument selection
- Computationally efficient



Technical Details

Outline

1. Introduction
2. Estimator
3. Asymptotic Results (Technical parts)
4. Simulation
5. An application: Wage Inequality in the United States

Model

$$
\begin{aligned}
Y_1^* &= X'\beta_0(U), \\
Y_2 &= \max\{Y_2^*, 0\} = \max\{X'\gamma_0(V), 0\}, \\
D &= 1\{Y_2^* > 0\}, \\
Y_1 &= DY_1^* \qquad (1)
\end{aligned}
$$

- Y_1^* and Y_2^* are log-wage level and hours worked
- D indicates labor force participation
- X are determinants of both levels, such as working experience, education, number of children and, etc

Correcting for Selection Bias

Identification Challenge

- Model 1 depends on two sources of unobserved heterogeneity:
  - Latent outcome rank U
  - Percentile V
- **Bias source**: Dependence between U and V causes sample selection bias
- **Solution**: Control the dependence structure **(copula functions)**

Identification

Selection Bias Correction

$$
\begin{aligned}
& P(Y_1 < X'\beta_0(u) | Y_2^* > X'\gamma_0(v), X = x) \\
=& P(Y_1^* < X'\beta_0(u) | X'\gamma_0(V) > X'\gamma_0(v), X = x) \\
=& \frac{C(u, 1, \rho_0) - C(u, v, \rho_0)}{1 - v} \\
\equiv& G(u, v, \rho_0),
\end{aligned} \tag{2}
$$

Given any pair of $u, v \in \mathcal{U}$ and for observations with $X'\gamma(v) > 0$

Interpretation

- $C(\cdot)$: Copula function capturing **dependence**
- ρ_0: Dependence parameter
- $G(u, v, \rho_0)$: Identified function for estimation

Conditional vs. Unconditional Moments

AB's approach

- Identification via conditional moment:
  - Method-of-Moment $E(\varepsilon|X) = 0$
- Estimation by its sample average
  - $\min \frac{1}{n}\sum_{i=1}^{n} \varepsilon_i \cdot g(X_i)$
  - Requires choosing instruments $g(X)$
  - Limited information:
    - X, X^2, $p(X)$, etc
    - Arbitrary selection
    - Inefficient: Uses only selected moments

Our approach

- Identification via unconditional moments:
  - Nonlinear Least Square $\min E(Y - m(X))^2$
- Estimation by its sample average
  - $\min \frac{1}{n}\sum_{i=1}^{n}(Y_i - m(X_i))^2$
  - No instrument selection
  - Full information:
    - No arbitrary choices
    - Efficient: Uses infinite moments

Back

Identification

Identification via unconditional moments

By the monotonicity of the cumulative distribution function, let

$$u = G^{-1}(\tau, v, \rho_0), \tag{3}$$

$$\begin{aligned}
&E[1\{Y_1 < X'\beta_0(u)\}|X, Y_2 > X'\gamma_0(\tilde{v}) > 0] \\
=&\frac{P(U < G^{-1}(\tau_m, v, \rho_0), V > \tilde{v}|X)}{P(V > \tilde{v}|X)} \\
=&\frac{C(G^{-1}(\tau_m, v, \rho_0), 1, \rho_0) - C(G^{-1}(\tau_m, v, \rho_0), \tilde{v}, \rho_0)}{1 - \tilde{v}} \\
=&G\left(G^{-1}(\tau_m, v, \rho_0), \tilde{v}, \rho_0\right),
\end{aligned} \tag{4}$$

with $X'\gamma(\tilde{v}) > 0$ by choosing some v, $\tilde{v}$ and a finite gird $\tau_1 < \tau_2 < ... < \tau_M$ on $\mathcal{U}$.

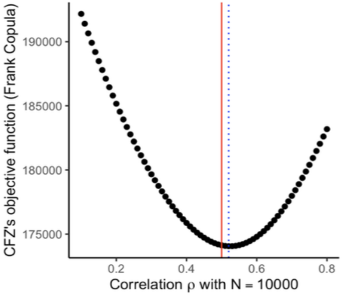
Estimation

Final Step: estimate the copula parameter ρ_0

we get $\hat{\rho}$ via

$$\min_{\rho \in \mathbb{R}} \int_{\tilde{v}} \int_{v} \sum_{m=1}^{M} \sum_{i=1}^{n} \left[1\{Y_{1i} < X_i' \hat{b}(\tau_m, v)\} - G^* \left(\tau_m, v, \tilde{v}, \rho\right) \right]^2$$

$$1\{Y_{2i} > X_i' \hat{\gamma}(\tilde{v}) > 0\} dv d\tilde{v}.$$



Estimation

First Step (selection equation): estimate parameters $\gamma_0(v)$

In model 1, we have

$$Y_2 = \max\{Y_2^*, 0\} = \max\{X'\gamma_0(V), 0\}$$
$$D = 1\{Y_2^* > 0\}$$

Due to the non-convex nature of the optimization problem in censored quantile regression, our model is based on Chen's (*JOE*, 2018). Define $\tilde{\gamma}(v_0)$ as the solution to

$$\min_{\gamma \in \mathbb{R}^p} \frac{1}{n} \sum_{i=1}^{n} \rho_{v_0}(Y_{2i} - X_i'\gamma) + \frac{\lambda}{n} \|\gamma\|_{1,n}, \tag{5}$$

and $\tilde{\gamma}(v_{l+1})$ as the solution to

$$\min_{\gamma \in \mathbb{R}^p} \frac{1}{n} \sum_{i=1}^{n} 1\{X_i'\tilde{\gamma}(v_l) > \delta_n\} \rho_{v_{l+1}}(Y_{2i} - X_i'\gamma) + \frac{\lambda}{n} \|\gamma\|_{1,n}, \tag{6}$$

for $v_{L_n} < \ldots < v_1 < v_0 \in \mathcal{U}$.

Estimation

First Step (selection equation): estimate parameters $\gamma_0(v)$

Addressing Estimation Bias

- Sparsity: Only a small subset of variables truly matter.
  Define the sparse supports as

$$\mathbb{S}_\gamma(v) = \text{support}(\gamma_0(v)) = \{j \in \{1, ..., p\} : |\gamma_{0j}(v)| > 0\},$$

  having at most $s^\gamma \leq n/\log(n(p+1))$ nonzero components respectively for all $v \in \mathcal{U}$.

- Hard-thresholding: Set coefficients below threshold to zero.
  Define the hard-thresholded estimators as

$$\hat{\gamma}_j(v) = \tilde{\gamma}_j(v) 1\{|\tilde{\gamma}_j(v)| > \varepsilon^\gamma\},$$

  for some $\varepsilon^\gamma \geq 0$, where $j = 1, .., p$ and $v \in \mathcal{U}$.

Back

Estimation

Second Step (outcome equation): estimate parameters $\beta_0(u)$

In model 1, we have

$$Y_1^* = X'\beta_0(U),$$
$$Y_1 = DY_1^*$$

We proposed $\tilde{b}(\tau, v)$ by solving

$$\min_{b \in \mathbb{R}^p} \frac{1}{n} \sum_{i=1}^{n} \rho_\tau(Y_{1i} - X_i'b) 1\{Y_{2i} > X_i'\tilde{\gamma}(v) > 0\} + \frac{\lambda}{n} \|b\|_{1,n}, \tag{7}$$

for any $\tau, v \in \mathcal{U}$.
Define the hard-thresholded estimators as

$$\hat{b}_j(\tau, v) = \tilde{b}_j(\tau, v) 1\{|\tilde{b}_j(\tau, v)| > \varepsilon^b\},$$

for some $\varepsilon^b \geq 0$ where $j = 1, .., p$ and $\tau, v \in \mathcal{U}$.

Outline

Asymptotic Results

Theorem (Convergence Rate)

Under general conditions, the ℓ_1-penalized quantile regression in the first two steps is uniformly consistent at the near-oracle rate $\sqrt{\frac{s^i \log((p+1)n)}{n}}$, where $i \in \{\gamma, b\}$.

Key conditions:

- **Sparsity:**

$$\sup_{v \in \mathcal{U}} \|\gamma(v)\|_0 \leq s^\gamma \quad \text{and} \quad \sup_{u \in \mathcal{U}} \|\beta(u)\|_0 \leq s^b$$

- **Restricted eigenvalue:**

$$\inf_{v \in \mathcal{U}} \inf_{\substack{\Delta_\gamma \neq 0 \\ \Delta_\gamma \in \mathcal{A}_v}} \frac{\Delta_\gamma' E(1\{X_i \gamma_0(v) > \delta_0\} X_i X_i') \Delta_\gamma}{\|\Delta_\gamma\|^2} = \sigma^2_{\min} > 0$$

$$\inf_{v \in \mathcal{U}} \inf_{\substack{\Delta_b \neq 0 \\ \Delta_b \in \mathcal{B}_\tau}} \frac{\Delta_b' E\left(1\{Y_{2i} > X_i' \gamma_0(v) > 0\} X_i X_i'\right) \Delta_b}{\|\Delta_b\|^2} = \delta^2_{\min} > 0$$

for all i.

Asymptotic Results

Theorem (Hard-thresholding)

Let $r^\gamma = \sup_{v\in\mathcal{U}} \|\tilde{\gamma}(v) - \gamma(v)\|$. Assume $\inf_{v\in\mathcal{U}} \min_{j\in\mathbb{S}_\gamma(v)} |\gamma_j(v)| > c^\gamma$, where $r^\gamma < c^\gamma$, then

$$\mathbb{S}_\gamma(v) := \mathrm{support}(\gamma(v)) \subseteq \tilde{\mathbb{S}}_\gamma(v) := \mathrm{support}(\tilde{\gamma}(v)) \quad \text{for all } v \in \mathcal{U}. \tag{8}$$

Moreover, the hard-thresholded estimator $\hat{\gamma}(v)$, defined for any $\varepsilon^\gamma \geq 0$ by

$$\hat{\gamma}_j(v) = \tilde{\gamma}_j(v) 1\{|\tilde{\gamma}_j(v)| > \varepsilon^\gamma\}, \quad v \in \mathcal{U}, j = 1, \ldots, p, \tag{9}$$

provided that ε^γ is chosen such that $r^\gamma < \varepsilon^\gamma < c^\gamma - r^\gamma$, satisfies

$$\mathrm{support}(\hat{\gamma}(v)) = \mathbb{S}_\gamma(v) \quad \text{for all } v \in \mathcal{U}.$$

Asymptotic Results

Theorem

Under general assumptions, then $\hat{\rho}$ is consistent for ρ_0, and is asymptotically normal:

$$\sqrt{n}(\hat{\rho} - \rho_0) \to^d N(0, \Sigma_\rho).$$

Outline

Monte Carlo simulations

Performance of estimator $\hat{\rho}$ in normal settings

DGP

$$
\begin{aligned}
Y_1^* &= \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \sigma(X) \cdot \Phi^{-1}(U) \\
Y_2^* &= \gamma_0 + \gamma_1 X_1 + \gamma_2 X_2 + \Phi^{-1}(V) \\
D &= 1\{Y_2^* > 0\} \\
Y_2 &= \max(Y_2^*, 0) \\
Y_1 &= D Y_1^*
\end{aligned}
$$

- $X_1 \sim 0.5 \cdot N_1$, $X_2 \sim U[0,2]$ and N_1 are independent standard normal variables
- $\beta_0 = \beta_1 = \beta_2 = 1, \gamma_0 = 0.2, \gamma_1 = \gamma_2 = 1$
- U and V are uniform on $(0,1)$, with their dependence modelled through different copulas, and both are independent of X
- $\sigma(X) = 1$ for the homoscedastic case and $\sigma(X) = 1 + 0.5 \cdot X_1$ for the heteroscedastic case

Monte Carlo simulations

Performance of estimator $\hat{\rho}$ in normal settings

Frank copula V.S. Gaussian copula (homoscedastic case)

Param	True	Bias	RMSE	SD
Censoring $\approx$ 17%				
		$n = 300$		
τ	0.7000	-0.0212	0.0480	0.0430
		$n = 600$		
τ	0.7000	-0.0138	0.0332	0.0296
		$n = 300$		
τ	-0.7000	0.0163	0.0447	0.0420
		$n = 600$		
τ	-0.7000	0.0123	0.0332	0.0302

Frank copula

Param	True	Bias	RMSE	SD
Censoring $\approx$ 17%				
		$n = 300$		
ρ	0.7000	-0.0291	0.0768	0.0710
		$n = 600$		
ρ	0.7000	-0.0198	0.0510	0.0466
		$n = 300$		
ρ	-0.7000	0.0088	0.0707	0.0702
		$n = 600$		
ρ	-0.7000	0.0101	0.0469	0.0462

Gaussian copula

Monte Carlo simulations

Performance of estimator $\hat{\rho}$ in normal settings

Frank copula V.S. Gaussian copula (heteroscedastic case)

Param	True	Bias	RMSE	SD
Censoring $\approx$ 17%				
		$n = 300$		
τ	0.7000	-0.0225	0.0480	0.0429
		$n = 600$		
τ	0.7000	-0.0144	0.0332	0.0303
		$n = 300$		
τ	-0.7000	0.0181	0.0458	0.0427
		$n = 600$		
τ	-0.7000	0.0129	0.0332	0.0307

Frank copula

Param	True	Bias	RMSE	SD
Censoring $\approx$ 17%				
		$n = 300$		
ρ	0.7000	-0.0287	0.0768	0.0712
		$n = 600$		
ρ	0.7000	-0.0211	0.0520	0.0474
		$n = 300$		
ρ	-0.7000	0.0096	0.0721	0.0718
		$n = 600$		
ρ	-0.7000	0.0109	0.0479	0.0463

Gaussian copula

Monte Carlo simulations

Performance of estimator $\hat{\rho}$ in normal settings

Frank Copula: Varying dependence strengths

Param	True	Bias	RMSE	SD
Censoring Rate $\approx$ 17%				
		$n = 300$		
τ	0.3000	-0.0215	0.0900	0.0875
		$n = 600$		
τ	0.3000	-0.0108	0.0686	0.0679
		$n = 300$		
τ	0.5000	-0.0202	0.0678	0.0651
		$n = 600$		
τ	0.5000	-0.0165	0.0490	0.0461
		$n = 300$		
τ	0.7000	-0.0212	0.0480	0.0430
		$n = 600$		
τ	0.7000	-0.0138	0.0332	0.0296
		$n = 300$		
τ	0.9000	-0.0214	0.0265	0.0160
		$n = 600$		
τ	0.9000	-0.0154	0.0173	0.0088

Monte Carlo simulations

Performance of all estimators in high-dimensional settings

DGP

$$
\begin{aligned}
Y_1^* &= 1 + X_1 + X_2 + X_3 + \sigma(X) \cdot \Phi^{-1}(U) \\
Y_2^* &= 2.5 + X_1 + X_2 + X_3 + Z_1 + \Phi^{-1}(V) \\
D &= 1\{Y_2^* > 0\} \\
Y_2 &= \max(Y_2^*, 0) \\
Y_1 &= DY_1^*
\end{aligned}
$$

- The number of covariates is set to $p = 500$
- All covariates are independently drawn from a standard normal distribution
- The choice of penalty level λ follows Belloni and Chernozhukov (*AOS*, 2011)

Monte Carlo simulations

Performance of estimator $\hat{\gamma}(v)$ in high-dimensional settings

Frank copula (Kendall's $\tau = 0.7$, homoscedastic case)

τ	Param	True	$n=1000$ Bias	SD	RMSE	$n=2000$ Bias	SD	RMSE
0.30	γ_0	1.9756	-0.0055	0.0510	0.0512	0.0002	0.0353	0.0352
	γ_1	1.0000	-0.0544	0.0537	0.0764	-0.0383	0.0413	0.0563
	γ_2	1.0000	-0.0322	0.0617	0.0696	-0.0265	0.0430	0.0504
	γ_3	1.0000	-0.0308	0.0625	0.0696	-0.0211	0.0470	0.0514
	γ_4	1.0000	-0.0544	0.0562	0.0782	-0.0419	0.0411	0.0586
0.50	γ_0	2.5000	-0.0184	0.0458	0.0493	-0.0083	0.0314	0.0325
	γ_1	1.0000	-0.0481	0.0496	0.0691	-0.0350	0.0353	0.0497
	γ_2	1.0000	-0.0267	0.0574	0.0632	-0.0205	0.0399	0.0448
	γ_3	1.0000	-0.0234	0.0577	0.0622	-0.0188	0.0408	0.0449
	γ_4	1.0000	-0.0482	0.0513	0.0704	-0.0374	0.0404	0.0550
0.70	γ_0	3.0244	-0.0130	0.0438	0.0456	-0.0007	0.0308	0.0307
	γ_1	1.0000	-0.0537	0.0540	0.0762	-0.0408	0.0355	0.0540
	γ_2	1.0000	-0.0346	0.0579	0.0674	-0.0261	0.0393	0.0471
	γ_3	1.0000	-0.0310	0.0567	0.0646	-0.0271	0.0400	0.0483
	γ_4	1.0000	-0.0560	0.0523	0.0766	-0.0417	0.0379	0.0563
0.90	γ_0	3.7816	0.1834	0.0587	0.1925	0.2133	0.0377	0.2166
	γ_1	1.0000	-0.1739	0.0695	0.1872	-0.1669	0.0495	0.1741
	γ_2	1.0000	-0.1545	0.0759	0.1721	-0.1407	0.0553	0.1511
	γ_3	1.0000	-0.1463	0.0713	0.1627	-0.1443	0.0543	0.1541
	γ_4	1.0000	-0.1826	0.0686	0.1950	-0.1681	0.0484	0.1749

Monte Carlo simulations

Performance of estimator $\hat{\beta}(u)$ in high-dimensional settings

Frank copula (Kendall's $\tau = 0.7$, homoscedastic case)

			$n = 1000$			$n = 2000$		
τ	Param	True	Bias	SD	RMSE	Bias	SD	RMSE
0.3	β_1	1.00	-0.0645	0.0783	0.1013	-0.0711	0.0551	0.0899
	β_2	1.00	-0.0727	0.0818	0.1094	-0.0601	0.0589	0.0841
	β_3	1.00	-0.0985	0.0793	0.1264	-0.1044	0.0564	0.1186
0.5	β_1	1.00	-0.0587	0.0756	0.0956	-0.0626	0.0511	0.0808
	β_2	1.00	-0.0659	0.0766	0.1010	-0.0564	0.0568	0.0800
	β_3	1.00	-0.0903	0.0689	0.1135	-0.0938	0.0524	0.1074
0.7	β_1	1.00	-0.0550	0.0813	0.0981	-0.0541	0.0546	0.0768
	β_2	1.00	-0.0537	0.0826	0.0984	-0.0511	0.0595	0.0784
	β_3	1.00	-0.0823	0.0717	0.1091	-0.0816	0.0544	0.0980
0.9	β_1	1.00	-0.0472	0.1060	0.1160	-0.0447	0.0719	0.0845
	β_2	1.00	-0.0433	0.1082	0.1164	-0.0420	0.0779	0.0884
	β_3	1.00	-0.0728	0.0941	0.1189	-0.0675	0.0708	0.0977

Monte Carlo simulations

Performance of estimator $\hat{\beta}(u)$ in high-dimensional settings

Frank copula (Kendall's $\tau = 0.7$, homoscedastic case)

			$n = 1000$			$n = 2000$		
τ	Param	True	Bias	SD	RMSE	Bias	SD	RMSE
0.3	β_1	1.00	-0.0645	0.0783	0.1013	-0.0711	0.0551	0.0899
	β_2	1.00	-0.0727	0.0818	0.1094	-0.0601	0.0589	0.0841
	β_3	1.00	-0.0985	0.0793	0.1264	-0.1044	0.0564	0.1186
0.5	β_1	1.00	-0.0587	0.0756	0.0956	-0.0626	0.0511	0.0808
	β_2	1.00	-0.0659	0.0766	0.1010	-0.0564	0.0568	0.0800
	β_3	1.00	-0.0903	0.0689	0.1135	-0.0938	0.0524	0.1074
0.7	β_1	1.00	-0.0550	0.0813	0.0981	-0.0541	0.0546	0.0768
	β_2	1.00	-0.0537	0.0826	0.0984	-0.0511	0.0595	0.0784
	β_3	1.00	-0.0823	0.0717	0.1091	-0.0816	0.0544	0.0980
0.9	β_1	1.00	-0.0472	0.1060	0.1160	-0.0447	0.0719	0.0845
	β_2	1.00	-0.0433	0.1082	0.1164	-0.0420	0.0779	0.0884
	β_3	1.00	-0.0728	0.0941	0.1189	-0.0675	0.0708	0.0977

Monte Carlo simulations

Performance of estimator $\hat{\beta}(u)$ in high-dimensional settings

Frank copula (Kendall's $\tau = 0.7$, homoscedastic case)

			$n = 1000$			$n = 2000$		
τ	Param	True	Bias	SD	RMSE	Bias	SD	RMSE
	Kendall's τ	0.7000	-0.0351	0.0412	0.0212	-0.0200	0.02646	0.0163
0.3	β_0	0.4756	0.1974	0.0767	0.2118	0.1501	0.0667	0.1642
	β_1	1.0000	-0.0421	0.0743	0.0853	-0.0310	0.0598	0.0673
	β_2	1.0000	-0.0368	0.0787	0.0868	-0.0278	0.0660	0.0715
	β_3	1.0000	-0.0455	0.0743	0.0871	-0.0366	0.0596	0.0699
0.5	β_0	1.0000	0.0702	0.0606	0.0927	0.0477	0.0483	0.0679
	β_1	1.0000	-0.0385	0.0580	0.0695	-0.0286	0.0462	0.0542
	β_2	1.0000	-0.0287	0.0625	0.0687	-0.0219	0.0491	0.0537
	β_3	1.0000	-0.0404	0.0552	0.0684	-0.0355	0.0444	0.0568
0.7	β_0	1.5244	0.0232	0.0534	0.0582	0.0195	0.0389	0.0435
	β_1	1.0000	-0.0326	0.0545	0.0635	-0.0267	0.0389	0.0471
	β_2	1.0000	-0.0217	0.0609	0.0646	-0.0166	0.0430	0.0460
	β_3	1.0000	-0.0412	0.0523	0.0665	-0.0327	0.0386	0.0505
0.9	β_0	2.2816	0.0128	0.0606	0.0618	0.0128	0.0422	0.0440
	β_1	1.0000	-0.0259	0.0687	0.0733	-0.0244	0.0481	0.0538
	β_2	1.0000	-0.0253	0.0792	0.0830	-0.0143	0.0524	0.0543
	β_3	1.0000	-0.0315	0.0663	0.0733	-0.0268	0.0492	0.0559

Monte Carlo simulations

Performance of estimator $\hat{\beta}(u)$ in high-dimensional settings

Frank copula (Kendall's $\tau = 0.7$, heteroscedastic case)

τ	Param	True	$n = 1000$ Bias	SD	RMSE	$n = 2000$ Bias	SD	RMSE
	Kendall's τ	0.7000	-0.0482	0.0529	0.0223	-0.0316	0.0361	0.0173
0.3	β_0	0.4756	0.1489	0.0903	0.1741	0.1020	0.0833	0.1316
	β_1	0.7378	0.1323	0.0740	0.1515	0.1270	0.0615	0.1411
	β_2	1.0000	-0.0531	0.0896	0.1041	-0.0440	0.0774	0.0889
	β_3	1.0000	-0.0664	0.0840	0.1070	-0.0571	0.0707	0.0908
0.5	β_0	1.0000	0.0410	0.0666	0.0781	0.0233	0.0545	0.0592
	β_1	1.0000	0.0281	0.0600	0.0661	0.0293	0.0466	0.0549
	β_2	1.0000	-0.0385	0.0686	0.0786	-0.0312	0.0533	0.0617
	β_3	1.0000	-0.0517	0.0623	0.0809	-0.0458	0.0462	0.0650
0.7	β_0	1.5244	0.0089	0.0553	0.0560	0.0087	0.0401	0.0409
	β_1	1.2622	-0.0212	0.0560	0.0598	-0.0145	0.0368	0.0395
	β_2	1.0000	-0.0271	0.0616	0.0672	-0.0202	0.0415	0.0461
	β_3	1.0000	-0.0459	0.0542	0.0710	-0.0378	0.0383	0.0538
0.9	β_0	2.2816	0.0097	0.0609	0.0616	0.0091	0.0439	0.0447
	β_1	1.6408	-0.0476	0.0631	0.0790	-0.0401	0.0433	0.0590
	β_2	1.0000	-0.0250	0.0755	0.0794	-0.0150	0.0487	0.0509
	β_3	1.0000	-0.0336	0.0675	0.0753	-0.0283	0.0474	0.0551

Outline

Data

- American Community Survey (ACS) from 2001 to 2023
- a 10% stratified random sample is drawn, where stratification is based on state, race, and education level
- The final sample consists of 2,706,867 observations across the 2001-2023 period.

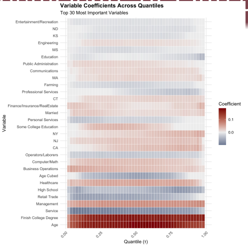
Results

The set of control variables X includes age, age squared, age cubed, three educational attainment dummies, 50 state dummies, a marital status dummy, 8 race category dummies, 14 occupation dummies, and 18 industry dummies.

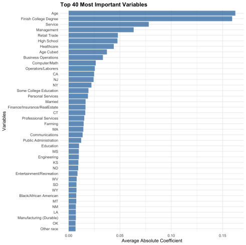
The degree of Selection

Year	Male	Female
2001	-0.164	-0.045
2002	-0.150	-0.060
2003	-0.150	-0.030
2004	-0.135	-0.105
2005	-0.120	-0.030
2006	-0.120	-0.045
2007	-0.105	0.000
2008	-0.105	-0.015
2009	-0.075	-0.015
2010	-0.120	-0.015
2011	-0.075	-0.015
2012	-0.105	0.000
2013	-0.105	-0.015
2014	-0.090	0.000
2015	-0.075	-0.015
2016	-0.090	-0.030
2017	-0.090	-0.030
2018	-0.075	-0.030
2019	-0.090	-0.015
2020	-0.090	-0.045
2021	-0.090	-0.075
2022	-0.090	-0.060
2023	-0.075	-0.045

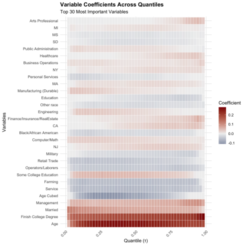
Results: Coefficients of Outcome Equations



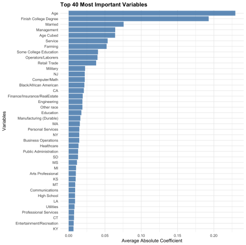
(a) Female: coefficients across quantiles



(b) Female: top 40 important variables



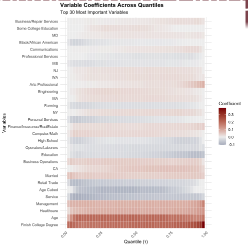
(c) Male: coefficients across quantiles



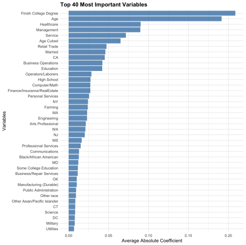
(d) Male: top 40 important variables

Selection-corrected coefficients for the year 2001.

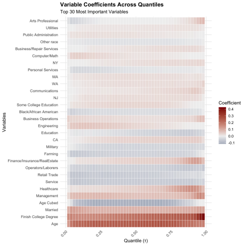
Results: Coefficients of Outcome Equations



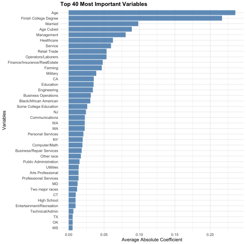
(a) Female: coefficients across quantiles



(b) Female: top 40 important variables



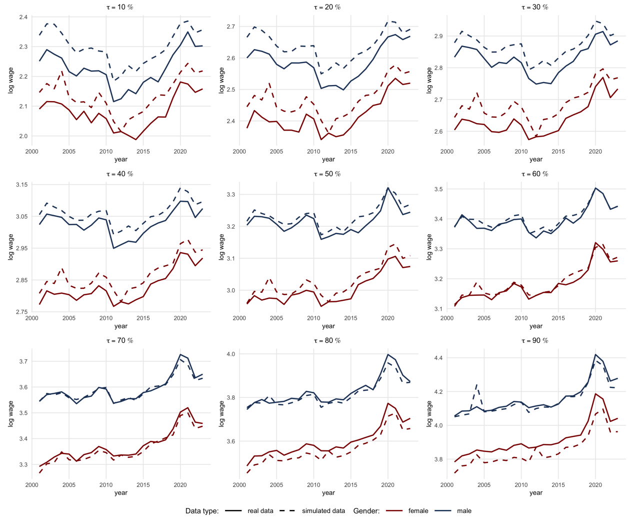
(c) Male: coefficients across quantiles



(d) Male: top 40 important variables

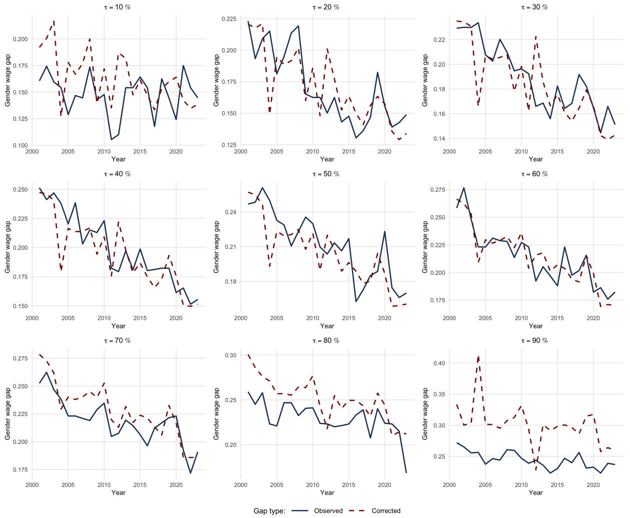
Selection-corrected coefficients for the year 2023.

Results: Selection-corrected Wage



Log-wage across quantiles (2001 - 2023)

Results: Wage inequality



Gender wage gaps across quantiles (2001 - 2023)

Thank You!

Questions Welcome

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Job Talk Q&A: Feel free to use the shared Google Doc for questions